CMUC Online Seminar on Algebra, Logic and Topology

University of Coimbra

Algebraic geometry and tangent categories

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Abstract

Tangent categories, first defined by Jiri Rosicky, are a categorical structure which encompasses not only ordinary differential geometry, but also synthetic differential geometry and notions of differentiation coming from computer science. In this talk, I'll discuss how the category of affine schemes is a tangent category, and how its tangent category structure relates to ideas in algebraic geometry. I'll also briefly discuss how the notion of a "differential bundle" in a tangent category gives a common generalization of (smooth) vector bundles and modules over a ring. This talk is based on joint work with JS Lemay.